SOFTWARE ENGINEERING

1. Software Engineering is defined as the application of a systematic, disciplined, and quantifiable approach to:

A) Hardware development
B) Software development, operation, and maintenance
C) Computer networking
D) Data storage
Answer: B

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2. The term software crisis emerged in the:

A) 1940s
B) 1950s
C) 1960s
D) 1990s
Answer: C

---

3. Which of the following is a fundamental component of software?

A) Programs
B) Documentation
C) Data structures
D) All of the above
Answer: D

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4. Which characteristic differentiates software from hardware?

A) Software is physical
B) Software does not wear out
C) Software is manufactured
D) Software requires electricity
Answer: B

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5. The primary objective of software engineering is to produce software that is:

A) Expensive
B) Reliable and efficient
C) Complex
D) Hardware dependent
Answer: B

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6. A software process refers to:

A) The physical assembly of computers
B) A set of activities used to develop and maintain software
C) A hardware testing method
D) A programming language
Answer: B

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7. Which of the following is a generic software engineering activity?

A) Specification
B) Development
C) Validation
D) All of the above
Answer: D

---

8. The activity that defines what the system should do is called:

A) Coding
B) Requirement engineering
C) Maintenance
D) Debugging
Answer: B

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9. Software validation ensures that:

A) The software is free from errors
B) The software meets user requirements
C) The program runs faster
D) Hardware compatibility exists
Answer: B

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10. The modification of software after delivery is known as:

A) Testing
B) Maintenance
C) Implementation
D) Verification
Answer: B

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11. Which of the following is NOT a generic framework activity in software engineering?

A) Planning
B) Modeling
C) Manufacturing
D) Deployment
Answer: C

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12. The IEEE definition of software engineering emphasizes:

A) Random programming
B) Structured and measurable development methods
C) Hardware engineering
D) Database management
Answer: B

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13. Which type of software manages computer hardware resources?

A) System software
B) Application software
C) Embedded software
D) Scientific software
Answer: A

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14. Embedded software is mainly used in:

A) Operating systems
B) Word processors
C) Hardware devices and control systems
D) Database servers
Answer: C

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15. The term legacy system refers to:

A) Newly developed software
B) Outdated software still in use
C) Temporary programs
D) Experimental software
Answer: B

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16. The process of checking whether the software is built correctly according to specifications is called:

A) Verification
B) Maintenance
C) Coding
D) Installation
Answer: A

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17. Software that performs specific tasks for users is called:

A) System software
B) Application software
C) Utility software
D) Firmware
Answer: B

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18. The major challenge in software engineering projects is:

A) Hardware availability
B) Managing complexity and changing requirements
C) Internet connectivity
D) Processor speed
Answer: B

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19. A software process model that follows a linear and sequential approach is called:
A) Spiral Model
B) Agile Model
C) Waterfall Model
D) RAD Model
Answer: C) Waterfall Model

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20. In the Waterfall model, which phase comes immediately after system design?
A) Coding
B) Testing
C) Maintenance
D) Deployment
Answer: A) Coding

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21. The Spiral Model mainly combines which two models?
A) Waterfall and Prototyping
B) Iterative and Risk Analysis
C) Agile and RAD

D) V-Model and Incremental
Answer: B) Iterative and Risk Analysis

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22. Which process model is risk-driven and focuses heavily on risk analysis?
A) Spiral Model
B) Waterfall Model
C) Incremental Model
D) Prototype Model
Answer: A) Spiral Model

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23. In the Prototyping model, the prototype is mainly developed to:
A) Replace the final system
B) Understand and refine user requirements
C) Reduce testing effort
D) Increase coding speed
Answer: B) Understand and refine user requirements

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24. Which software process model delivers the system in small functional parts called increments?
A) Incremental Model
B) Waterfall Model
C) V-Model
D) Spiral Model
Answer: A) Incremental Model

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25. The V-Model is also known as:
A) Verification and Validation Model
B) Visual Model
C) Virtual Model
D) Variable Model
Answer: A) Verification and Validation Model

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26. In the V-Model, each development phase is associated with:
A) Maintenance phase
B) Testing phase
C) Design phase
D) Coding phase
Answer: B) Testing phase

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27. Which model emphasizes rapid development using reusable components and tools?
A) RAD Model
B) Spiral Model
C) Waterfall Model
D) V-Model
Answer: A) RAD Model

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28. The Agile model focuses primarily on:
A) Heavy documentation
B) Customer collaboration and iterative delivery
C) Fixed requirements
D) Sequential development
Answer: B) Customer collaboration and iterative delivery

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29. Which software process model is most suitable when requirements are unclear at the beginning?
A) Prototyping Model
B) Waterfall Model
C) V-Model
D) Big Bang Model
Answer: A) Prototyping Model

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30. The Big Bang Model is typically used for:
A) Large and complex projects
B) Small experimental projects
C) Government systems

D) Safety-critical systems
Answer: B) Small experimental projects

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31. In the Incremental Model, each increment goes through which phases?
A) Design only
B) Requirements only
C) Complete development cycle
D) Testing only
Answer: C) Complete development cycle

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32. Which model is considered rigid and difficult to modify once a phase is completed?
A) Agile Model
B) Waterfall Model
C) Spiral Model
D) Incremental Model
Answer: B) Waterfall Model

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33. The Spiral Model consists of multiple loops, where each loop represents:
A) Coding phase
B) Development cycle
C) Testing phase
D) Maintenance stage
Answer: B) Development cycle

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34. Which process model is best suited for large and high-risk projects?
A) Waterfall Model
B) Spiral Model
C) Big Bang Model
D) RAD Model
Answer: B) Spiral Model

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35. The primary objective of Agile methodology is to:
A) Eliminate documentation
B) Deliver working software frequently
C) Avoid customer interaction
D) Increase project cost
Answer: B) Deliver working software frequently

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36. Which process model integrates systematic testing with each development phase?
A) Prototyping Model
B) V-Model
C) Big Bang Model
D) RAD Model
Answer: B) V-Model

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37. Agile software development primarily emphasizes:
A) Detailed documentation
B) Sequential development
C) Customer collaboration
D) Fixed project scope
Answer: C) Customer collaboration

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38. The Agile Manifesto was introduced in:
A) 1995
B) 2001
C) 2005
D) 2010
Answer: B) 2001

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39. Which of the following is a core value of the Agile Manifesto?
A) Processes and tools over individuals
B) Working software over comprehensive documentation
C) Contract negotiation over collaboration

D) Following a strict plan over responding to change
Answer: B) Working software over comprehensive documentation

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40. Agile development divides the project into small cycles called:
A) Modules
B) Iterations
C) Phases
D) Pipelines
Answer: B) Iterations

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41. In Agile methodology, the list of project requirements is called the:
A) Sprint backlog
B) Product backlog
C) Requirement log
D) Task board
Answer: B) Product backlog

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42. Which Agile framework is widely used for managing iterative development?
A) Scrum
B) Waterfall
C) V-Model
D) Spiral
Answer: A) Scrum

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43. In Scrum, the person responsible for maximizing product value is called the:
A) Scrum Master
B) Product Owner
C) Project Manager
D) System Analyst
Answer: B) Product Owner

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44. The Scrum meeting held daily to track progress is called:
A) Sprint Planning
B) Sprint Review
C) Daily Stand-up
D) Retrospective
Answer: C) Daily Stand-up

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45. In Agile development, a sprint typically lasts:
A) 1–4 weeks
B) 6–8 months
C) 2 years
D) 1 day
Answer: A) 1–4 weeks

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46. Which Agile practice involves writing tests before coding?
A) Pair Programming
B) Continuous Integration
C) Test-Driven Development
D) Refactoring
Answer: C) Test-Driven Development

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47. The Agile principle that focuses on improving code structure without changing functionality is:
A) Refactoring
B) Integration
C) Deployment
D) Prototyping
Answer: A) Refactoring

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48. Which Agile technique involves two programmers working together on the same code?
A) Incremental coding

B) Pair programming
C) Code sharing
D) Dual programming
Answer: B) Pair programming

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49. Continuous integration in Agile means:
A) Writing all code at once
B) Integrating code frequently into a shared repository
C) Testing only at the end
D) Integrating documentation
Answer: B) Integrating code frequently into a shared repository

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50. In Scrum, the Scrum Master is responsible for:
A) Managing budget
B) Removing obstacles for the team
C) Writing all code
D) Approving requirements
Answer: B) Removing obstacles for the team

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51. The meeting where the team reflects on improvements after a sprint is called:
A) Sprint Review
B) Sprint Retrospective
C) Sprint Planning
D) Daily Scrum
Answer: B) Sprint Retrospective

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52. Which Agile framework focuses heavily on engineering practices like pair programming and TDD?
A) Extreme Programming (XP)
B) Scrum
C) Kanban

D) Lean
Answer: A) Extreme Programming (XP)

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53. In Agile, user stories are used to:
A) Replace coding
B) Describe features from the user's perspective
C) Document system architecture
D) Perform testing
Answer: B) Describe features from the user's perspective

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54. Agile development encourages:
A) Late testing only
B) Minimal customer involvement
C) Continuous feedback and improvement
D) Strict sequential phases
Answer: C) Continuous feedback and improvement

55. Requirement Engineering primarily deals with:
A) Software coding
B) Identifying, analyzing, and documenting system requirements
C) Testing the system
D) Maintaining the software
Answer: B) Identifying, analyzing, and documenting system requirements

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56. The process of collecting requirements from stakeholders is known as:
A) Requirement validation
B) Requirement elicitation
C) Requirement specification
D) Requirement management
Answer: B) Requirement elicitation

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57. Which document contains a detailed description of system requirements?
A) Software Design Document (SDD)

B) Software Requirement Specification (SRS)
C) Test Plan
D) User Manual
Answer: B) Software Requirement Specification (SRS)

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58. Requirements that describe system behavior and functionality are called:
A) Non-functional requirements
B) Functional requirements
C) User requirements
D) Technical requirements
Answer: B) Functional requirements

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59. Requirements that specify performance, security, or usability constraints are called:
A) Functional requirements
B) Non-functional requirements
C) System requirements
D) Operational requirements
Answer: B) Non-functional requirements

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60. Which technique involves structured interaction with stakeholders to gather requirements?
A) Interview
B) Compilation
C) Coding
D) Refactoring
Answer: A) Interview

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61. A graphical representation used to describe interactions between users and a system is called:
A) Flowchart
B) Use Case Diagram
C) Data Table
D) Structure Chart
Answer: B) Use Case Diagram

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62. Requirement analysis mainly focuses on:
A) Writing program code
B) Detecting conflicts and refining requirements
C) System deployment
D) Maintenance activities
Answer: B) Detecting conflicts and refining requirements

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63. The process of checking whether requirements reflect the real needs of stakeholders is called:
A) Requirement validation
B) Requirement design
C) Requirement coding
D) Requirement maintenance
Answer: A) Requirement validation

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64. A requirement that can be tested and verified is called:
A) Ambiguous requirement
B) Verifiable requirement
C) Complex requirement
D) Temporary requirement
Answer: B) Verifiable requirement

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65. The activity of organizing and documenting requirements in a standard format is known as:
A) Requirement specification
B) Requirement elicitation
C) Requirement verification
D) Requirement implementation
Answer: A) Requirement specification

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66. Which model is commonly used to represent system functions and data flow during requirement analysis?
A) Data Flow Diagram (DFD)

B) ER Diagram
C) Structure Chart
D) Control Chart
Answer: A) Data Flow Diagram (DFD)

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67. The process of managing requirement changes during development is called:
A) Requirement control
B) Requirement management
C) Requirement design
D) Requirement coding
Answer: B) Requirement management

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68. Which characteristic ensures that requirements are clear and have only one interpretation?
A) Ambiguity
B) Consistency
C) Unambiguity
D) Completeness
Answer: C) Unambiguity

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69. A requirement that can be traced to its origin is known as:
A) Traceable requirement
B) Abstract requirement
C) Logical requirement
D) Dynamic requirement
Answer: A) Traceable requirement

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70. Which stakeholder is primarily responsible for understanding and analyzing system requirements?
A) Software tester
B) System analyst
C) Network administrator
D) Hardware engineer
Answer: B) System analyst

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71. Prototyping in requirement engineering is mainly used to:
A) Finalize coding
B) Validate and clarify requirements
C) Reduce maintenance
D) Replace documentation
Answer: B) Validate and clarify requirements

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72. Which of the following is a quality attribute of a good SRS document?
A) Ambiguous
B) Incomplete
C) Consistent
D) Redundant
Answer: C) Consistent

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73. Software Project Management mainly focuses on:
A) Writing program code
B) Planning, organizing, and controlling software development activities
C) Testing software only
D) Designing hardware systems
Answer: B) Planning, organizing, and controlling software development activities

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74. The first phase in software project management is:
A) Project planning
B) Project execution
C) Project initiation
D) Project monitoring
Answer: C) Project initiation

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75. The document that defines project objectives, scope, and stakeholders is called:
A) Project Charter
B) Test Plan
C) Design Document

D) Maintenance Report
Answer: A) Project Charter

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76. A graphical representation used to show project tasks and schedules is known as:
A) Data Flow Diagram
B) Gantt Chart
C) ER Diagram
D) State Diagram
Answer: B) Gantt Chart

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77. Which technique is used to estimate project cost and effort based on lines of code?
A) Agile estimation
B) COCOMO Model
C) Spiral estimation
D) V-Model estimation
Answer: B) COCOMO Model

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78. In project management, milestones represent:
A) Coding errors
B) Important project checkpoints
C) Hardware failures
D) Testing tools
Answer: B) Important project checkpoints

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79. The process of identifying potential problems that may affect a project is called:
A) Risk management
B) Cost estimation
C) Requirement analysis
D) Quality control
Answer: A) Risk management

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80. Which activity involves assigning tasks and responsibilities to team members?
A) Resource allocation
B) Requirement specification
C) System testing
D) Maintenance planning
Answer: A) Resource allocation

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81. The technique used to determine the longest path of activities in a project schedule is:
A) PERT
B) CPM
C) RAD
D) Agile
Answer: B) CPM

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82. Which project management tool represents task dependencies using a network diagram?
A) PERT Chart
B) Gantt Chart
C) Flowchart
D) Data Table
Answer: A) PERT Chart

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83. The process of comparing actual progress with planned progress is known as:
A) Project execution
B) Project monitoring
C) Project coding
D) Project documentation
Answer: B) Project monitoring

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84. Which factor mainly influences software project cost estimation?
A) Code complexity
B) Programming language
C) Team experience

D) All of the above
Answer: D) All of the above

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85. The critical path in project scheduling represents:
A) Shortest path in the project
B) Sequence of activities determining project duration
C) Optional project tasks
D) Testing sequence
Answer: B) Sequence of activities determining project duration

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86. Which activity ensures that the software product meets required quality standards?
A) Quality assurance
B) Requirement elicitation
C) Risk analysis
D) Resource allocation
Answer: A) Quality assurance

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87. The process of controlling modifications to project scope is called:
A) Scope management
B) Risk control
C) Resource management
D) Cost management
Answer: A) Scope management

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88. A person responsible for planning and managing a software project is called:
A) System Analyst
B) Software Tester
C) Project Manager
D) Database Administrator
Answer: C) Project Manager

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89. Which estimation technique is based on expert judgment and experience?
A) Algorithmic estimation
B) Expert judgment estimation
C) Parametric estimation
D) Automated estimation
Answer: B) Expert judgment estimation

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90. The final phase of project management where project activities are formally completed is called:
A) Project closure
B) Project testing
C) Project planning
D) Project design
Answer: A) Project closure

MCQs on Software Design

91. Software design is the process of:
A) Writing program code
B) Transforming requirements into a structured solution
C) Testing software
D) Maintaining software
Answer: B) Transforming requirements into a structured solution

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92. The main objective of software design is to:
A) Reduce testing time
B) Provide a blueprint for implementation
C) Eliminate documentation
D) Replace requirement analysis
Answer: B) Provide a blueprint for implementation

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93. Which design level focuses on the overall system architecture?
A) Detailed design
B) Architectural design
C) Interface design
D) Component design
Answer: B) Architectural design

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94. The design that describes internal logic and algorithms of components is called:
A) High-level design
B) Low-level design
C) System design
D) Conceptual design
Answer: B) Low-level design

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95. Which principle ensures that each module performs a single well-defined task?
A) High cohesion
B) Low cohesion
C) High coupling
D) Tight coupling
Answer: A) High cohesion

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96. The degree of interdependence between modules is called:
A) Cohesion
B) Coupling
C) Modularity
D) Abstraction
Answer: B) Coupling

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97. Which type of coupling occurs when modules share global data?
A) Data coupling
B) Common coupling
C) Control coupling
D) Content coupling
Answer: B) Common coupling

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98. The process of dividing a system into smaller manageable modules is known as:
A) Encapsulation
B) Modularity

C) Integration
D) Compilation
Answer: B) Modularity

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99. Which design concept focuses on hiding internal details and exposing only essential features?
A) Abstraction
B) Coupling
C) Cohesion
D) Decomposition
Answer: A) Abstraction

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100. The design principle that hides implementation details from users is called:
A) Encapsulation
B) Decomposition
C) Integration
D) Configuration
Answer: A) Encapsulation

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101. A diagram used to represent system components and their relationships is called:
A) Class Diagram
B) Flowchart
C) Data Table
D) Decision Tree
Answer: A) Class Diagram

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102. Which design method focuses on breaking a system into smaller subsystems?
A) Top-down design
B) Bottom-up design
C) Incremental design
D) Rapid design
Answer: A) Top-down design

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103. Which design approach starts with detailed components and integrates them into a complete system?
A) Top-down design
B) Bottom-up design
C) Structured design
D) Modular design
Answer: B) Bottom-up design

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104. A good software design should primarily aim for:
A) High coupling and low cohesion
B) Low coupling and high cohesion
C) High complexity
D) Large modules
Answer: B) Low coupling and high cohesion

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105. Interface design mainly focuses on:
A) Database structure
B) Interaction between system components or users
C) Coding techniques
D) Hardware configuration
Answer: B) Interaction between system components or users

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106. Which diagram represents dynamic interactions between objects over time?
A) Sequence Diagram
B) Class Diagram
C) ER Diagram
D) Structure Chart
Answer: A) Sequence Diagram

107. The concept of step-by-step refinement of a program design is known as:
A) Design decomposition
B) Stepwise refinement
C) Encapsulation
D) Abstraction
Answer: B) Stepwise refinement

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108. The design stage in software development comes after:
A) Coding
B) Requirement analysis
C) Testing
D) Maintenance
Answer: B) Requirement analysis

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109. Which design principle emphasizes representing only the essential features of an entity while hiding unnecessary details?

A. Abstraction
B. Coupling
C. Refactoring
D. Cohesion

Answer: A

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110. A software module is said to have high cohesion when it:

A. Depends heavily on other modules
B. Performs a single, well-defined task
C. Contains many unrelated functions
D. Has no interface

Answer: B

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111. Which of the following best describes coupling in software design?

A. The internal strength of a module
B. The number of comments in code
C. The degree of interdependence between modules
D. The size of a class

Answer: C

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112. Stepwise refinement in software design means:

A. Starting from code and moving to requirements
B. Ignoring details in early design
C. Replacing architecture with testing
D. Developing the design in increasing levels of detail

Answer: D

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113. Which of the following is most desirable in a good software design?

A. High cohesion and low coupling
B. High coupling and low cohesion
C. High complexity and high dependency
D. Low modularity and high redundancy

Answer: A

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114. Functional independence of modules is achieved when modules have:

A. Low cohesion and high coupling
B. High cohesion and low coupling
C. Low cohesion and low abstraction
D. High inheritance and no interfaces

Answer: B

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115. Which design model focuses on data structures, architecture, interfaces, and components?

A. Testing model
B. Deployment model
C. Design model
D. Maintenance model

Answer: C

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116. Information hiding means that a module should:

A. Avoid all documentation
B. Expose all internal data
C. Hide implementation details behind interfaces
D. Share all variables globally

Answer: C

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117. Which UML diagram is most suitable for showing the static relationships among classes?

A. Sequence diagram
B. Use case diagram
C. Activity diagram
D. Class diagram

Answer: D

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118. The main benefit of modularity in software design is:

A. Easier understanding, testing, and maintenance
B. More duplicated code
C. Increased dependency
D. Removal of all design decisions

Answer: A

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119. Which of the following is a behavioral design pattern?

A. Adapter
B. Singleton

C. Observer
D. Bridge

Answer: C

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120. The Factory pattern is mainly used to:

A. Separate interface from implementation
B. Create objects without exposing exact creation logic
C. Reduce cohesion
D. Increase code duplication

Answer: B

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121. A design is considered maintainable if it:

A. Is difficult to understand
B. Uses tightly coupled modules
C. Can be modified easily with minimal side effects
D. Avoids all documentation

Answer: C

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122. Which of the following best describes top-down design?

A. Designing detailed low-level modules first
B. Designing from the overall system toward detailed components
C. Starting with testing cases
D. Writing code before analysis

Answer: B

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123. Bottom-up design starts with:

A. User manuals
B. High-level requirements only
C. Detailed reusable components
D. Deployment diagrams

Answer: C

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124. A software interface specification mainly defines:

A. Employee roles
B. Communication rules between components
C. Testing budgets
D. Coding syntax of a language

Answer: B

---

125. Which of the following is a key sign of poor design?

A. Clear separation of responsibilities
B. Consistent naming
C. Reusable modules
D. Excessive dependency among modules

Answer: D

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126. Refactoring improves software design by:

A. Changing functionality for users
B. Improving internal structure without changing behavior
C. Removing all abstraction
D. Eliminating documentation

Answer: B

---

127. Software architecture primarily defines:

A. The detailed syntax of code
B. The high-level structure of a system
C. Test cases for modules
D. User training procedures

Answer: B

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128. In software architecture, a component is:

A. A physical network cable
B. A user manual
C. A modular unit with specific responsibilities
D. A project milestone

Answer: C

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129. Connectors in architecture are responsible for:

A. Writing code
B. Linking hardware ports
C. Storing user data
D. Enabling interaction among components

Answer: D

---

130. Which architecture style organizes software into layers of services?

A. Layered architecture
B. Repository architecture
C. Pipe-and-filter architecture
D. Event-driven architecture

Answer: A

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131. In a layered architecture, each layer usually:

A. Ignores the layer above it
B. Uses services of the layer below and serves the layer above
C. Communicates with all layers directly
D. Has no defined role

Answer: B

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132. Which architectural style is most appropriate for centralized shared data?

A. Event-driven
B. Client-server
C. Repository
D. Layered

Answer: C

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133. The client-server architecture separates systems into:

A. Data objects and classes
B. Processes and threads
C. Service requesters and service providers
D. Interfaces and adapters

Answer: C

---

134. Which quality attribute refers to the ability to handle increased workload?

A. Portability
B. Usability
C. Maintainability
D. Scalability

Answer: D

---

135. Microservices architecture is characterized by:

A. One monolithic code base only
B. Large tightly coupled modules
C. Small independently deployable services
D. No communication between services

Answer: C

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136. Which of the following is a major advantage of software architecture?

A. It eliminates all coding
B. It helps satisfy quality attributes early
C. It replaces requirements engineering
D. It avoids documentation

Answer: B

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137. The pipe-and-filter architectural style is best suited for:

A. Sequential transformation of data
B. Centralized control of UI
C. Role-based security only
D. Distributed databases only

Answer: A

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138. Event-driven architecture is particularly useful when:

A. Components must respond to asynchronous events
B. Only one module exists
C. There is no user interaction
D. Data is never shared

Answer: A

---

139. Which document commonly records major architectural decisions?

A. Test case report
B. Configuration log
C. Software Architecture Document
D. Payroll sheet

Answer: C

---

140. An architectural view is used to:

A. Replace source code
B. Represent a system from a particular stakeholder concern
C. Test performance directly
D. Define coding standards only

Answer: B

---

141. Which UML diagram is most suitable for representing physical deployment nodes?

A. Deployment diagram
B. Class diagram
C. Sequence diagram
D. State diagram

Answer: A

---

142. Architectural degradation occurs when:

A. Architecture becomes clearer with time
B. Continuous uncontrolled changes reduce structural quality
C. Testing is automated
D. Security policies are strengthened

Answer: B

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143. Which architecture style is commonly used in web applications with presentation, logic, and data tiers?

A. Pipe-and-filter
B. Three-tier architecture
C. Event bus architecture
D. Blackboard architecture

Answer: B

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144. Architectural evaluation is performed to:

A. Measure typing speed
B. Check compliance with hardware vendors only
C. Assess whether the architecture meets quality and functional needs
D. Eliminate all future changes

Answer: C

---

145. The main objective of user interface design is to:

A. Maximize internal code complexity
B. Create effective and satisfying interaction for users
C. Avoid feedback
D. Replace system testing

Answer: B

---

146. Which UI principle ensures similar elements behave in similar ways?

A. Consistency
B. Encapsulation
C. Refactoring
D. Coupling

Answer: A

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147. Usability mainly refers to:

A. Processor speed
B. Memory usage
C. Ease of learning and using the interface
D. Number of code files

Answer: C

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148. Which of the following is a good example of feedback in UI?

A. Hidden errors
B. A progress indicator during upload
C. Removing navigation options
D. Silent failure of commands

Answer: B

---

149. Error prevention in UI design aims to:

A. Increase user confusion
B. Delay system response
C. Minimize chances of user mistakes
D. Hide warnings

Answer: C

---

150. A wireframe is:

A. A complete coded system
B. A rough visual layout of interface elements
C. A test execution report
D. A network diagram

Answer: B

---

151. Prototyping in UI design helps to:

A. Finalize deployment architecture
B. Test and refine interface ideas early
C. Replace all documentation
D. Eliminate user feedback

Answer: B

---

152. Which of the following best supports readability in an interface?

A. Tiny fonts and no spacing
B. Random alignment
C. Clear labels and proper visual spacing
D. Excessive animation

Answer: C

---

153. Accessibility in UI design means:

A. Designing only for expert users
B. Supporting users with diverse abilities
C. Using mouse input only
D. Avoiding keyboard shortcuts

Answer: B

---

154. Which UI element is most suitable for selecting one option from several mutually exclusive options?

A. Check box
B. Text area
C. Radio button
D. Progress bar

Answer: C

---

155. Visual hierarchy helps users by:

A. Making all items equally prominent
B. Hiding important actions
C. Organizing attention toward the most important elements first
D. Removing labels

Answer: C

---

156. Which design approach focuses on users' needs, goals, and limitations?

A. Hardware-centered design
B. User-centered design
C. Technology-first design
D. Code-centered design

Answer: B

---

157. Which of the following is an example of direct manipulation in UI?

A. Dragging a file into a folder
B. Editing configuration code manually
C. Restarting the system after every command
D. Writing only command scripts

Answer: A

---

158. A responsive interface is one that:

A. Works only on desktop screens
B. Avoids mobile devices
C. Adapts to different screen sizes and devices
D. Uses fixed layouts only

Answer: C

---

159. Which of the following is a poor UI design practice?

A. Immediate system feedback
B. Consistent button placement
C. Clear grouping of related form fields
D. Overloading one screen with too much information

Answer: D

---

160. Confirmation dialogs are especially useful for:

A. Frequent harmless actions
B. Destructive actions such as delete
C. Every single mouse movement
D. Displaying advertisements

Answer: B

---

161. Minimalist design recommends:

A. Showing every possible control on the home screen
B. Including only essential information and controls
C. Hiding all instructions
D. Avoiding structure

Answer: B

---

162. Usability testing is mainly conducted to:

A. Measure bandwidth
B. Verify payroll systems
C. Observe whether users can effectively perform tasks
D. Replace integration testing

Answer: C

---

163. Coding standards are mainly intended to:

A. Increase ambiguity
B. Improve readability and consistency
C. Replace testing
D. Remove modularity

Answer: B

---

164. Which of the following is the best indicator of maintainable code?

A. Duplicated logic everywhere
B. Long functions with unclear names
C. Consistent style and meaningful identifiers
D. Global variables in every module

Answer: C

---

165. Refactoring is the process of:

A. Adding new features only
B. Improving internal code structure without changing external behavior
C. Deleting documentation
D. Changing requirements

Answer: B

---

166. Defensive programming encourages developers to:

A. Assume all inputs are valid
B. Avoid validation
C. Anticipate unexpected situations and handle them safely
D. Remove exception handling

Answer: C

---

167. Which of the following supports code reuse the most?

A. Hard-coded values
B. Generic functions and well-defined classes
C. Repeated copy-paste blocks
D. Hidden dependencies

Answer: B

---

168. Dead code is code that:

A. Is highly reusable
B. Is never executed or no longer needed
C. Improves performance automatically
D. Has many comments

Answer: B

---

169. A version control system is used to:

A. Track and manage source code changes
B. Replace compilers
C. Design networks
D. Remove all bugs automatically

Answer: A

---

170. Which of the following is a common sign of poor implementation?

A. Short focused functions
B. Meaningful variable names
C. Deeply nested duplicated logic
D. Automated builds

Answer: C

---

171. Exception handling is primarily used to:

A. Prevent documentation
B. Manage runtime errors gracefully
C. Increase coupling
D. Replace testing

Answer: B

---

172. Which of the following best describes portability in coding?

A. Code can run on different platforms with little change
B. Code has no functions
C. Code is locked to one machine
D. Code uses only global data

Answer: A

---

173. Magic numbers should be avoided because they:

A. Clarify intent immediately
B. Make maintenance and understanding harder
C. Improve modularity
D. Reduce the need for named constants in all cases

Answer: B

---

174. Encapsulation helps implementation by:

A. Exposing all internal details
B. Protecting data and hiding implementation details
C. Increasing duplication
D. Avoiding interfaces

Answer: B

---

175. Which practice is most effective for early detection of coding issues?

A. Deployment testing only
B. Code review
C. User training
D. Acceptance sign-off

Answer: B

---

176. A function with too many parameters may indicate:

A. Good modularity
B. Strong cohesion
C. Possible design weakness and low readability
D. Perfect abstraction

Answer: C

---

177. The main role of a compiler is to:

A. Translate source code into machine or intermediate code
B. Test usability
C. Draw UML diagrams
D. Estimate project risk

Answer: A

---

178. Which of the following best supports secure coding?

A. Trusting all input data
B. Validating and sanitizing external inputs
C. Storing secrets in plain text
D. Ignoring boundary cases

Answer: B

---

179. Low coupling in implementation means:

A. Modules are highly dependent on each other
B. Modules interact minimally through clear interfaces
C. Code is stored in one file only
D. Variables are global

Answer: B

---

180. An IDE typically provides:

A. Only source code storage
B. Only testing reports
C. Editing, debugging, and build support
D. Only network simulation

Answer: C

---

181. The fundamental purpose of software testing is to:

A. Guarantee software is defect-free
B. Find defects and verify expected behavior
C. Replace design activities
D. Avoid user feedback

Answer: B

---

182. Unit testing is primarily concerned with:

A. Entire organizations
B. The interaction between integrated subsystems
C. Individual components or functions
D. Acceptance by customers

Answer: C

---

183. Integration testing focuses on:

A. Interaction between combined modules
B. Only user manuals
C. Only hardware installation
D. Team management

Answer: A

---

184. System testing verifies:

A. Only the database
B. The complete integrated system against requirements
C. Only source code comments
D. Project scope only

Answer: B

---

185. Acceptance testing is mainly used to determine whether:

A. The software meets business and user needs
B. All code paths are executed
C. Developers are satisfied
D. The architecture is layered

Answer: A

---

186. Regression testing is performed to:

A. Verify old functionality still works after changes
B. Test hardware speed
C. Replace integration testing
D. Train users

Answer: A

---

187. Black-box testing is based on:

A. Internal code structure
B. External behavior and expected outputs
C. Class inheritance only
D. Developer salary records

Answer: B

---

188. White-box testing emphasizes:

A. UI attractiveness
B. Contract compliance
C. Internal logic, branches, and paths
D. User preference surveys

Answer: C

---

189. Boundary value analysis is useful because errors often occur:

A. In architecture documents
B. Near the limits of input ranges
C. In employee records
D. In deployment rooms

Answer: B

---

190. Equivalence partitioning reduces testing effort by:

A. Dividing inputs into representative classes
B. Testing every input individually
C. Ignoring invalid data
D. Replacing test cases with assumptions

Answer: A

---

191. Smoke testing is generally used to:

A. Check basic stability of a build before detailed testing
B. Measure fire safety
C. Replace stress testing
D. Verify all use cases completely

Answer: A

---

192. Stress testing evaluates system behavior:

A. Under ideal conditions only
B. Under extreme or beyond-normal load
C. Only during code review
D. Only for user manuals

Answer: B

---

193. Performance testing mainly examines:

A. Code comments
B. Visual design only
C. Speed, responsiveness, and stability under workload
D. Documentation structure

Answer: C

---

194. Retesting is performed to:

A. Confirm a specific defect has been fixed
B. Remove requirements
C. Replace validation
D. Estimate cost

Answer: A

---

195. Beta testing is usually carried out:

A. Internally by developers only
B. By selected users in a real environment
C. Before coding begins
D. Without functional software

Answer: B

---

196. A test oracle is:

A. A random guess
B. A method for determining expected test outcomes
C. A compiler plugin
D. A database server

Answer: B

---

197. Code coverage measures:

A. Percentage of source code executed by tests
B. Number of users trained
C. Quality of documentation fonts
D. Size of the project budget

Answer: A

---

198. Which statement about testing is most accurate?

A. Testing proves absence of all defects
B. Testing is needed only after coding
C. Testing can reveal defects, not prove perfection
D. Testing replaces quality assurance

Answer: C

---

199. Software maintenance refers to:

A. Activities performed after delivery to correct or improve software
B. Only initial coding
C. Only project estimation
D. Only documentation printing

Answer: A

---

200. Corrective maintenance is performed to:

A. Adapt software to new environments
B. Fix faults discovered after delivery
C. Improve performance only
D. Reduce future risks only

Answer: B

---

201. Adaptive maintenance is needed when:

A. The software must work in a changed environment
B. New coding begins
C. A project charter is updated
D. User training ends

Answer: A

---

202. Perfective maintenance generally aims to:

A. Remove all modules
B. Fix syntax errors only
C. Improve functionality, performance, or usability
D. Change project ownership only

Answer: C

---

203. Preventive maintenance seeks to:

A. Reduce future maintenance problems
B. Add customer-requested features only
C. Replace system testing
D. Convert all software to hardware

Answer: A

---

204. Software evolution describes:

A. Gradual change of software over time
B. Physical wear of disks
C. Automatic deletion of code
D. Ending all software support

Answer: A

---

205. Legacy systems are often difficult to evolve because they:

A. Always use modern technology
B. Usually have excellent documentation
C. May be built on outdated platforms with poor structure
D. Never change

Answer: C

---

206. Reverse engineering is used to:

A. Understand an existing system from code and artifacts
B. Avoid maintenance
C. Delete architecture documents
D. Replace testing tools

Answer: A

---

207. Re-engineering mainly focuses on:

A. Improving internal structure of an existing system
B. Eliminating all functions
C. Writing new requirements only
D. Building hardware devices

Answer: A

---

208. Technical debt is best defined as:

A. Money spent on software tools
B. Future cost caused by quick but poor technical decisions
C. License fees for frameworks
D. The cost of documentation printing

Answer: B

---

209. Impact analysis helps maintainers to:

A. Check how a proposed change affects other parts of the system
B. Remove all old code immediately
C. Skip regression testing
D. Replace design reviews

Answer: A

---

210. Which factor most increases maintenance difficulty?

A. Clear architecture
B. Consistent coding standards
C. Tightly coupled poorly documented code
D. High cohesion

Answer: C

---

211. Regression testing during maintenance is important because:

A. Changes may introduce side effects in existing functionality
B. Old features never break
C. Testing after release is unnecessary
D. Maintenance affects only documentation

Answer: A

---

212. Program comprehension means:

A. Understanding the existing code before modifying it
B. Deleting old versions
C. Measuring response time
D. Replacing analysts

Answer: A

---

213. Software aging refers to:

A. Hardware failure
B. Increased difficulty in maintaining software due to accumulated changes
C. Expiration of source code
D. Automatic loss of functionality with time only

Answer: B

---

214. Configuration management during maintenance helps control:

A. Changes to versions and related artifacts
B. User interface colors
C. Office attendance
D. Processor temperature

Answer: A

---

215. Maintainability is the ease with which software can be:

A. Marketed and sold
B. Understood, corrected, adapted, and enhanced
C. Printed on paper
D. Executed without memory

Answer: B

---

216. Software Quality Assurance is primarily concerned with:

A. Ensuring processes and products satisfy quality standards
B. Only writing source code
C. Only testing final builds
D. Only hiring staff

Answer: A

---

217. SQA differs from testing because SQA focuses more on:

A. Defect prevention through process quality
B. Writing acceptance criteria only
C. Running compilers
D. Preparing UI prototypes

Answer: A

---

218. A quality audit is performed to:

A. Check process compliance with standards and plans
B. Generate code automatically
C. Train users
D. Replace inspections

Answer: A

---

219. The quality assurance plan mainly defines:

A. Payroll rules
B. Quality procedures, standards, and responsibilities
C. Programming syntax
D. Network topology

Answer: B

---

220. Verification asks:

A. Are we building the right product?
B. Is the software profitable?
C. Are we building the product right?
D. Can we skip testing?

Answer: C

---

221. Validation asks:

A. Are we building the right product?
B. Are we coding efficiently?
C. Are we using enough comments?
D. Are we following budget only?

Answer: A

---

222. Which of the following is a preventive SQA activity?

A. Fixing a production failure
B. Conducting formal design reviews
C. Restarting services
D. Archiving code

Answer: B

---

223. Peer reviews support quality assurance by:

A. Detecting defects early in work products
B. Replacing all tests
C. Avoiding collaboration
D. Preventing future maintenance forever

Answer: A

---

224. A software metric is used to:

A. Remove all defects automatically
B. Quantify attributes of product or process
C. Replace configuration management
D. Approve budgets only

Answer: B

---

225. Defect density is a measure of:

A. Defects relative to software size
B. Developer salaries
C. Number of interface colors
D. User login frequency

Answer: A

---

226. Root cause analysis aims to:

A. Rename files
B. Identify underlying causes of defects
C. Increase schedule pressure
D. Eliminate documentation

Answer: B

---

227. Standards are important in SQA because they:

A. Provide a consistent basis for evaluation and work practices
B. Remove the need for expertise
C. Eliminate all bugs
D. Apply only to hardware

Answer: A

---

228. Process improvement in SQA aims to:

A. Reduce discipline
B. Continuously improve development practices
C. Replace architecture
D. Avoid measurement

Answer: B

---

229. Which of the following is a product quality attribute?

A. Reliability
B. Office rent
C. Team hierarchy
D. Contract duration

Answer: A

---

230. Statistical quality control in software uses:

A. Random guesses
B. Quantitative data and measurements
C. Only interviews
D. Only source code formatting

Answer: B

---

231. A checklist during inspection helps by:

A. Guiding systematic review and reducing omission
B. Replacing standards
C. Avoiding documentation
D. Limiting reviewer participation

Answer: A

---

232. SQA is most effective when it is:

A. Applied only at project end
B. Integrated throughout the software life cycle
C. Limited to testing teams only
D. Used only after failures occur

Answer: B

---

233. A software metric is best defined as:

A. A qualitative opinion about software
B. A quantitative measure of a software attribute
C. A replacement for reviews
D. A project contract

Answer: B

---

234. Which of the following is a product metric?

A. Defect density
B. Average effort per week
C. Cost variance
D. Team size

Answer: A

---

235. Which of the following is a process metric?

A. Number of defects per KLOC
B. Cyclomatic complexity
C. Mean time to fix defects
D. Number of classes

Answer: C

---

236. Metrics are useful in software engineering because they:

A. Eliminate all uncertainty
B. Support objective analysis and decision-making
C. Replace all management judgment
D. Prevent all change requests

Answer: B

---

237. LOC stands for:

A. Level of coupling
B. Logic of coding
C. Lines of Code
D. Length of class

Answer: C

---

238. Function Point Analysis measures software size based on:

A. Hardware devices
B. Delivered functionality to the user
C. Only source code lines
D. Only number of programmers

Answer: B

---

239. Cyclomatic complexity measures:

A. User satisfaction
B. Project budget
C. The number of independent paths in a program
D. Number of interface windows

Answer: C

---

240. A higher cyclomatic complexity generally suggests:

A. Easier testing and simpler logic
B. Lower code risk automatically
C. Greater structural complexity and testing effort
D. Better reusability in all cases

Answer: C

---

241. Mean Time Between Failures (MTBF) is associated with:

A. Reliability measurement
B. Coding style
C. Data normalization
D. Interface responsiveness only

Answer: A

---

242. Mean Time To Repair (MTTR) measures:

A. The average time to correct and restore service after failure
B. The average number of lines per module
C. The total number of test cases
D. The number of users per release

Answer: A

---

243. Which metric is most useful for maintainability assessment?

A. Number of sales reports
B. Coupling and complexity measures
C. User age statistics
D. Screen brightness

Answer: B

---

244. A valid software metric should be:

A. Ambiguous
B. Difficult to interpret
C. Clearly defined and repeatable
D. Based only on personal opinion

Answer: C

---

245. Measurement helps project control by:

A. Providing data on progress, quality, and effort
B. Eliminating the need for planning
C. Avoiding all risk
D. Removing customers from decisions

Answer: A

---

246. Goal-Question-Metric (GQM) is used to:

A. Link metrics to organizational goals through questions
B. Replace testing strategies
C. Write code faster
D. Standardize hardware selection

Answer: A

---

247. Which of the following is a size-oriented metric?

A. Function points
B. Lines of Code
C. Defect removal efficiency
D. MTTR

Answer: B

---

248. Defect Removal Efficiency (DRE) evaluates:

A. How effectively defects are removed before release
B. Hardware utilization
C. Number of UI screens
D. Data transfer speed

Answer: A

---

249. Which of the following is most likely a misuse of metrics?

A. Using them to guide improvement
B. Using them in context with judgment
C. Using one metric alone to make all decisions
D. Tracking trends over time

Answer: C

---

250. Software Configuration Management primarily deals with:

A. Controlling and tracking changes to software artifacts
B. Designing database schemas
C. Training users
D. Measuring processor performance

Answer: A

---

251. A configuration item is:

A. Any software artifact placed under control
B. Only a source code file
C. Only a hardware device
D. Only a design diagram

Answer: A

---

252. Version control systems help teams by:

A. Tracking changes and supporting collaboration
B. Eliminating all merge conflicts
C. Replacing testing
D. Preventing all defects

Answer: A

---

253. A baseline in SCM is:

A. A temporary draft that changes every minute
B. An approved reference version used for future comparison
C. A deleted file
D. A user account list

Answer: B

---

254. Change control ensures that:

A. All changes are ignored
B. Only approved and evaluated changes are implemented
C. Every developer changes anything anytime
D. Source code is never modified

Answer: B

---

255. Configuration status accounting is concerned with:

A. Recording and reporting the status of configuration items and changes
B. User training schedules
C. Interface colors
D. Budget estimation only

Answer: A

---

256. Configuration auditing is used to:

A. Check whether configuration items conform to requirements and records
B. Rewrite requirements
C. Replace design reviews
D. Increase coupling

Answer: A

---

257. Branching in version control is useful for:

A. Creating isolated lines of development
B. Deleting release history
C. Avoiding collaboration
D. Preventing merges forever

Answer: A

---

258. Merging in SCM means:

A. Splitting a project into modules
B. Combining changes from different branches
C. Removing baselines
D. Archiving documentation only

Answer: B

---

259. A build in SCM refers to:

A. A compiled and assembled version of the software
B. A user acceptance report
C. A project contract
D. A risk checklist only

Answer: A

---

260. Release management is concerned with:

A. Planning and delivering approved versions to users
B. Drawing class diagrams
C. Writing only unit tests
D. Hiring developers

Answer: A

---

261. Which of the following is a major benefit of SCM?

A. Better traceability and control over software changes
B. No need for documentation
C. Elimination of all bugs
D. Removal of reviews

Answer: A

---

262. Check-in and check-out operations are associated with:

A. Configuration repositories
B. Network hardware
C. User registration systems
D. Payroll management

Answer: A

---

263. An SCM repository stores:

A. Controlled versions of source and related artifacts
B. Only executable files
C. Only user passwords
D. Only architecture diagrams

Answer: A

---

264. Traceability in SCM helps to:

A. Link requirements, changes, versions, and related artifacts
B. Hide design decisions
C. Reduce documentation
D. Increase ambiguity

Answer: A

---

265. Which of the following is most likely to cause configuration problems?

A. Uncontrolled parallel changes
B. Version labeling
C. Baseline approval
D. Change logs

Answer: A

---

266. A tag or label in version control is typically used to:

A. Mark a specific version or release point
B. Delete files permanently
C. Replace branches
D. Prevent documentation

Answer: A

---

267. Software risk management is concerned with:

A. Identifying, analyzing, and responding to project risks
B. Writing code only
C. Replacing architecture design
D. Avoiding all planning

Answer: A

---

268. A software risk is best defined as:

A. A guaranteed project failure
B. An uncertain event that may negatively affect objectives
C. A completed defect fix
D. A coding standard

Answer: B

---

269. Risk identification is the process of:

A. Discovering potential problems that may affect the project
B. Solving all risks immediately
C. Eliminating uncertainty completely
D. Avoiding all estimates

Answer: A

---

270. Risk analysis typically considers:

A. Only team size
B. Probability and impact
C. Only technical details
D. Coding style only

Answer: B

---

271. A risk register is used to:

A. Store project source code
B. Track identified risks and planned responses
C. Replace schedules
D. Record payroll data

Answer: B

---

272. Risk mitigation means:

A. Ignoring low risks
B. Reducing probability or impact of a risk
C. Accepting all risks
D. Deleting requirements

Answer: B

---

273. Which response is most suitable when a risk must not be allowed to occur?

A. Accept
B. Transfer
C. Avoid
D. Ignore

Answer: C

---

274. Risk monitoring means:

A. Watching identified risks and reassessing them over time
B. Testing only after release
C. Measuring CPU temperature
D. Freezing all project activity

Answer: A

---

275. Which of the following is a common technical risk?

A. Requirement volatility
B. Office relocation
C. Budget approval delay only
D. Staff vacation schedule only

Answer: A

---

276. Which of the following is a project risk?

A. Insufficient staff or unrealistic schedule
B. SQL syntax error
C. Button color mismatch
D. Data type conversion only

Answer: A

---

277. A contingency plan is prepared to:

A. Define what to do if a risk materializes
B. Replace the main schedule
C. Avoid documentation
D. Remove all uncertainty instantly

Answer: A

---

278. Risk exposure is commonly understood as:

A. The combination of likelihood and loss/impact
B. A project's number of developers
C. Testing effort only
D. A legal contract value

Answer: A

---

279. Risk prioritization helps managers:

A. Focus on the most significant risks first
B. Ignore all minor risks completely
C. Replace requirement analysis
D. Cancel testing activities

Answer: A

---

280. Which of the following is an external risk?

A. Changes in laws or regulations
B. Poor coding standards
C. Low module cohesion
D. Missing comments in code

Answer: A

---

281. Prototyping can reduce risk mainly by:

A. Clarifying requirements and technical feasibility
B. Increasing uncertainty
C. Replacing maintenance
D. Removing quality assurance

Answer: A

---

282. Which statement is most accurate about software risks?

A. Risks disappear after planning
B. Risks should be managed continuously throughout the project
C. Risks matter only at the end
D. Risks are always technical only

Answer: B

---

283. Software security engineering is primarily concerned with:

A. Building software that resists malicious attacks and misuse
B. Increasing interface colors
C. Replacing testing with encryption
D. Avoiding all user input

Answer: A

---

284. Which of the following best describes confidentiality?

A. Preventing unauthorized disclosure of information
B. Ensuring software is easy to modify
C. Making systems faster
D. Improving portability

Answer: A

---

285. Integrity in software security means:

A. Data is protected from unauthorized alteration
B. Data is compressed efficiently
C. Users are trained regularly
D. Screens are responsive

Answer: A

---

286. Availability refers to:

A. Ensuring systems and data are accessible when needed
B. Hiding all resources
C. Limiting all users permanently
D. Replacing backups

Answer: A

---

287. Authentication is the process of:

A. Verifying the identity of a user or system
B. Granting permissions automatically
C. Encrypting all traffic
D. Monitoring logs only

Answer: A

---

288. Authorization determines:

A. Who a user is
B. What an authenticated user is allowed to do
C. Whether data is encrypted
D. Whether software is portable

Answer: B

---

289. Which of the following is the best example of secure coding practice?

A. Trusting all input
B. Validating and sanitizing inputs
C. Storing passwords in plain text
D. Ignoring exceptions

Answer: B

---

290. SQL injection is an attack that targets:

A. Database queries built from unsafe input
B. User interface colors
C. Project budgets
D. Deployment diagrams

Answer: A

---

291. Cross-Site Scripting (XSS) mainly occurs when:

A. Untrusted input is included in web pages without proper handling
B. Databases are normalized
C. Passwords are hashed strongly
D. Firewalls are configured properly

Answer: A

---

292. Encryption is used to:

A. Convert data into a protected form unreadable without a key
B. Increase coupling
C. Replace authentication
D. Remove audit trails

Answer: A

---

293. Hashing is commonly used for:

A. Secure storage of passwords
B. Recoverable encryption of files
C. User interface rendering
D. Database indexing only

Answer: A

---

294. The principle of least privilege means:

A. Users should have only the minimum access necessary
B. Every user should be admin
C. Security logs are optional
D. Interfaces should be hidden

Answer: A

---

295. Threat modeling is used to:

A. Identify threats, vulnerabilities, and mitigations early
B. Replace testing
C. Write deployment code
D. Remove architecture

Answer: A

---

296. A vulnerability is:

A. A weakness that can be exploited
B. A successful attack only
C. A secure configuration
D. A backup copy

Answer: A

---

297. Multi-factor authentication improves security by:

A. Requiring more than one type of verification
B. Reducing password strength
C. Removing user identity checks
D. Disabling logs

Answer: A

---

298. Logging and auditing are important because they:

A. Support detection, investigation, and accountability
B. Eliminate need for prevention
C. Replace access control
D. Prevent all attacks automatically

Answer: A

---

299. Secure software design should address security:

A. Only after deployment
B. Only after user complaints
C. Throughout the software life cycle
D. Only during coding

Answer: C

---

300. Which of the following best summarizes software security engineering?

A. It is only about antivirus tools
B. It integrates security requirements, design, coding, and validation into development
C. It replaces software quality assurance
D. It focuses only on network hardware

Answer: B

WEB DEVELOPMENT

Q1. What is web development?
a) Only designing website layouts
b) The process of developing, building, and maintaining websites
c) Only writing JavaScript code
d) Managing web hosting servers

Answer: b

Q2. Which of the following is NOT a component of web development?
a) Web design
b) Database management
c) Computer hardware manufacturing
d) Programming

Answer: c

Q3. Front-end development primarily deals with:
a) Server-side logic
b) Database management
c) What users see and interact with in their browser
d) API integrations

Answer: c

Q4. Back-end development focuses on:
a) Colors and typography
b) Server-side logic, databases, and application functionality
c) HTML structure only
d) Browser compatibility

Answer: b

Q5. A full-stack developer works with:
a) Only front-end technologies

b) Only back-end technologies
c) Both front-end and back-end technologies
d) Only database management

Answer: c

Q6. What is a web browser?
a) A software application for accessing and viewing websites
b) A programming language
c) A database system
d) A web server

Answer: c

Q7. Which of the following is an example of a web browser?
a) Apache
b) MySQL
c) Google Chrome
d) Node.js

Answer: c

Q8. What is a website?
a) A single HTML file
b) A collection of related web pages under a common domain name
c) A database
d) A programming language

Answer: b

Q9. What is a web page?
a) A complete website
b) A single document or resource on the World Wide Web
c) A web server
d) A programming framework

Answer: b

Q10. Which language is fundamental for structuring content on the web?
a) JavaScript
b) Python
c) HTML
d) SQL

Answer: c

Q11. What percentage of websites currently use HTML?
a) Approximately 50%
b) Approximately 75%
c) Approximately 95.8%
d) Approximately 100%

Answer: c

Q12. The most recent version of HTML is:
a) HTML4
b) XHTML
c) HTML5
d) HTML 2010

Answer: c

Q13. What does CSS stand for?
a) Computer Style Sheets
b) Cascading Style Sheets
c) Creative Style System
d) Colorful Style Sheets

Answer: b

Q14. What is the primary purpose of JavaScript in web development?
a) Structuring content
b) Styling web pages

c) Adding interactivity and dynamic behavior
d) Managing databases

Answer: c

Q15. Which of the following is a front-end development framework?
a) Django
b) Laravel
c) React
d) Express.js

Answer: c

Q16. What does HTTP stand for?
a) HyperText Transfer Protocol
b) High Technology Transfer Protocol
c) HyperTransfer Text Protocol
d) HyperText Transmission Program

Answer: a

Q17. HTTP is a _____ protocol.
a) Stateful
b) Stateless
c) Connection-oriented only
d) Database

Answer: b

Q18. Which HTTP method is used to retrieve resources from a server?
a) POST
b) GET
c) PUT
d) DELETE

Answer: b

Q19. Which HTTP method is used to submit data to be processed to a server?
a) GET
b) FETCH
c) POST
d) SEND

Answer: c

Q20. What is a URL?
a) Universal Resource Locator
b) Uniform Resource Locator
c) Unified Resource Link
d) Universal Reference Link

Answer: b

Q21. In the URL "https://example.com/about", what is "https"?
a) The domain name
b) The protocol
c) The path
d) The query string

Answer: b

Q22. What does the HTTP status code "200 OK" indicate?
a) The request was successful
b) The page was not found
c) There was a server error
d) The request is forbidden

Answer: a

Q23. What does the HTTP status code "404" indicate?
a) Server error
b) Unauthorized access

c) Page not found
d) Request successful

Answer: c

Q24. What is a cookie in web development?
a) A small text file stored on the client's computer by the web server
b) A type of malware
c) A browser plugin
d) A database record

Answer: a

Q25. What is the purpose of cookies?
a) To store viruses
b) To remember user preferences and session information
c) To increase internet speed
d) To display ads

Answer: b

Q26. What is the difference between URI and URL?
a) They are exactly the same
b) A URL is a type of URI that specifies how to access a resource
c) A URI is a type of URL
d) They have no relationship

Answer: b

Q27. What is a web server?
a) A computer that hosts websites and serves web pages to clients
b) A web browser
c) A programming language
d) A database system

Answer: a

Q28. Which of the following is an example of a web server software?
a) Google Chrome
b) Apache
c) MySQL
d) JavaScript

Answer: b

Q29. What is the purpose of the "Host" header in an HTTP request?
a) To identify the user's operating system
b) To specify the domain name of the server
c) To encrypt the request
d) To set cookies

Answer: b

Q30. What is a client-server architecture?
a) A model where clients request services from servers
b) A model where all computers are equal
c) A single computer system
d) A database model

Answer: a

Q31. What is the default port for HTTP?
a) 80
b) 443
c) 21
d) 25

Answer: a

Q32. What is the default port for HTTPS?
a) 80
b) 443

c) 8080
d) 8443

Answer: b

Q33. What is the main difference between HTTP and HTTPS?
a) HTTPS is faster
b) HTTPS encrypts data using SSL/TLS
c) HTTPS uses a different port only
d) There is no difference

Answer: b

Q34. What is REST in web architecture?
a) A programming language
b) An architectural style for designing networked applications
c) A database system
d) A web server

Answer: b

Q35. Which of the following is a constraint of REST architecture?
a) Client-server architecture
b) Stateful interactions
c) Tight coupling
d) Single layer

Answer: a

Q36. What does HTML stand for?
a) HyperText Markup Language
b) High-Level Text Markup Language
c) HyperTransfer Markup Language
d) HyperText Modeling Language

Answer: a

Q37. Which tag is used to define an HTML document?
a) <body>
b) <head>
c) <html>
d) <document>

Answer: c

Q38. Where is the metadata of an HTML document contained?
a) <body> tag
b) <head> tag
c) <footer> tag
d) <main> tag

Answer: c

Q39. Which tag is used to set the title of an HTML document that appears in the browser tab?
a) <header>
b) <heading>
c) <title>
d) <caption>

Answer: c

Q40. Which is the correct sequence of tags to start a web page?
a) HTML, Body, Title, Head
b) HTML, Head, Title, Body
c) Head, Title, HTML, Body
d) HTML, Title, Head, Body

Answer: b

Q41. HTML tags are _____.
a) Case-sensitive

b) Not case-sensitive
c) Always uppercase
d) Always lowercase

Answer: b

Q42. Which tag is used for the largest heading in HTML?
a) <h6>
b) <h1>
c) <heading>
d) <header>

Answer: b

Q43. Which tag is used for the smallest heading in HTML?
a) <h1>
b) <h5>
c) <h6>
d) <h7>

Answer: c

Q44. Which tag is used to create a paragraph in HTML?
a) <para>
b) <p>
c) <paragraph>
d) <text>

Answer: b

Q45. Which tag is used to create a hyperlink?
a) <link>
b) <a>
c) <href>
d) <url>

Answer: b

Q46. What attribute is used to specify the destination URL in an anchor tag?
a) src
b) link
c) href
d) url

Answer: c

Q47. Which attribute opens a link in a new browser tab?
a) new="tab"
b) target="_blank"
c) open="new"
d) tab="new"

Answer: b

Q48. Which tag is used to insert an image in HTML?
a) <image>
b) <img>
c) <pic>
d) <picture>

Answer: b

Q49. What attribute specifies the image source in an <img> **tag?**
a) href
b) link
c) src
d) source

Answer: c

Q50. What is the purpose of the alt **attribute in an** <img> **tag?**
a) To specify image height
b) To provide alternative text if the image cannot be displayed

c) To add a tooltip
d) To specify image width

Answer: b

Q51. Which tag is used for an unordered list?
a) <ol>
b) <li>
c) <ul>
d) <list>

Answer: c

Q52. Which tag is used for an ordered list?
a) <ul>
b) <li>
c) <ol>
d) <order>

Answer: c

Q53. Which tag represents a list item?
a) <item>
b) <li>
c) <list-item>
d) <it>

Answer: b

Q54. Which tag is used to insert a line break?
a) <break>
b) <lb>
c)
d) <newline>

Answer: c

Q55. What is the difference between <b> **and** **tags?**
a) They are identical
b) <b> makes text bold, indicates strong importance
c) is deprecated
d) <b> is for links

Answer: b

Q56. Which tag is used to create a table row?
a) <tr>
b) <td>
c) <th>
d) <row>

Answer: a

Q57. Which tag is used to define a table cell?
a) <tr>
b) <td>
c) <th>
d) <cell>

Answer: b

Q58. Which tag is used to create a checkbox in HTML?
a) <checkbox>
b) <input type="check">
c) <input type="checkbox">
d) <check>

Answer: c

Q59. What is the difference between id **and** class **attributes?**
a) They are the same
b) id is unique per page, class can be used on multiple elements

c) class is unique, id can be reused
d) id is for CSS only

Answer: b

Q60. What is an empty tag in HTML?
a) A tag with no content that doesn't need a closing tag
b) A tag with no attributes
c) A tag that is deprecated
d) A tag that contains no text

Answer: a

Q61. Which of the following is an empty tag?
a)
b) <p>
c) <img>
d)

Answer: c

Q62. Which tag is used to add a comment in HTML?
a) <comment>...</comment>
b) <!-- ... -->
c) // ...
d) /* ... */

Answer: b

Q63. Which HTML5 tag is used for navigation links?
a) <navigation>
b) <nav>
c) <navigate>
d) <links>

Answer: b

Q64. Which HTML5 tag is used for the main content of a document?
a) <content>
b) <main>
c) <body>
d)

Answer: b

Q65. Which HTML5 tag is a semantic element for a self-contained composition?
a) <section>
b)
c) <article>
d) <piece>

Answer: c

Q66. What does CSS stand for?
a) Computer Style Sheets
b) Cascading Style Sheets
c) Creative Style System
d) Colorful Style Sheets

Answer: b

Q67. Which HTML tag is used to define an internal CSS style?
a) <css>
b) <style>
c) <script>
d) <styles>

Answer: b

Q68. Which HTML attribute is used to define inline CSS?
a) class
b) style

c) css
d) font

Answer: b

Q69. Which is the correct CSS syntax?
a) body {color: black;}
b) {body:color=black;}
c) body: {color black;}
d) {body;color:black;}

Answer: a

Q70. Which property is used to change the text color of an element?
a) font-color
b) text-color
c) color
d) foreground-color

Answer: c

Q71. Which CSS property controls the text size?
a) text-size
b) font-style
c) font-size
d) text-style

Answer: c

Q72. What is the correct CSS selector to select all paragraph elements?
a) .p
b) #p
c) p
d) *p

Answer: c

Q73. What does the # symbol represent in a CSS selector?
a) Class selector
b) ID selector
c) Element selector
d) Universal selector

Answer: b

Q74. What does the . symbol represent in a CSS selector?
a) ID selector
b) Class selector
c) Universal selector
d) Descendant selector

Answer: b

Q75. What does the * selector do in CSS?
a) Selects all elements with a class
b) Selects all elements in the document
c) Selects the first element
d) Selects elements by ID

Answer: b

Q76. What are the components of the CSS box model?
a) Content, padding, border, margin
b) Content, margin, outline, border
c) Text, padding, border, outline
d) Width, height, padding, margin

Answer: a

Q77. What is the difference between margin **and** padding**?**
a) They are the same
b) Margin is space inside the border, padding is outside

c) Margin is space outside the border, padding is inside
d) Margin is for block elements only

Answer: c

Q78. How do you center a block element horizontally using CSS?
a) center: yes;
b) align: center;
c) margin: 0 auto;
d) text-align: center;

Answer: c

Q79. Which CSS property is used to change the font of an element?
a) font-style
b) text-style
c) font-family
d) typeface

Answer: c

Q80. Which property is used to make text bold?
a) font: bold;
b) style: bold;
c) font-weight: bold;
d) text-decoration: bold;

Answer: c

Q81. What is the default value of the position **property?**
a) relative
b) absolute
c) fixed
d) static

Answer: d

Q82. What does position: relative **do?**
a) Positions element relative to the viewport
b) Positions element relative to its normal position
c) Positions element relative to the nearest positioned ancestor
d) Removes element from document flow

Answer: b

Q83. What does position: absolute **do?**
a) Positions element relative to its normal position
b) Positions element relative to the nearest positioned ancestor
c) Positions element relative to the document body
d) Keeps element fixed during scrolling

Answer: b

Q84. What does z-index **control?**
a) The zoom level of an element
b) The stacking order of elements
c) The index of elements in a list
d) The size of an element

Answer: b

Q85. Which CSS property is used to change the background color?
a) bgcolor
b) background-color
c) color
d) background

Answer: b

Q86. Which property is used to change the left margin of an element?
a) padding-left
b) margin-left

c) indent
d) left-margin

Answer: b

Q87. How do you select an element with id "header" in CSS?
a) .header
b) #header
c) header
d) *header

Answer: b

Q88. How do you select all elements with class "menu" in CSS?
a) .menu
b) #menu
c) menu
d) *menu

Answer: a

Q89. What does div p **selector mean?**
a) Selects all div elements inside a p
b) Selects all p elements inside a div
c) Selects div elements with a parent p
d) Selects p elements directly after a div

Answer: b

Q90. What does div > p **selector mean?**
a) Selects all p elements anywhere inside a div
b) Selects all p elements that are direct children of a div
c) Selects div elements that are inside p
d) Selects p elements that are siblings of a div

Answer: b

Q91. Which property is used to set the width of an element?
a) height
b) width
c) size
d) length

Answer: b

Q92. Which property is used to add space inside an element's border?
a) margin
b) padding
c) spacing
d) inset

Answer: b

Q93. Which CSS property controls the visibility of an element?
a) show
b) visible
c) display
d) view

Answer: c

Q94. What is the default value of the display **property for a** **?**
a) inline
b) block
c) inline-block
d) flex

Answer: b

Q95. What is the default value of the display **property for a** **?**
a) block
b) inline

c) inline-block
d) flex

Answer: b

Q96. What is responsive web design?
a) Designing websites that respond quickly
b) Creating websites that adapt to different screen sizes and devices
c) Websites with animations
d) Websites that only work on mobile

Answer: b

Q97. What is a media query in CSS?
a) A query to fetch media files
b) A CSS technique to apply styles based on device characteristics
c) A JavaScript function
d) A HTML attribute

Answer: b

Q98. What is the purpose of the viewport meta tag?
a) To set the page title
b) To control layout on mobile browsers
c) To add a favicon
d) To link CSS files

Answer: b

Q99. What does flex-direction: column **do in Flexbox?**
a) Arranges items horizontally
b) Arranges items vertically in a column
c) Reverses the order of items
d) Creates a grid layout

Answer: b

Q100. In Flexbox, what does justify-content **align?**
a) Items along the vertical axis
b) Items along the horizontal axis (main axis)
c) All items to the center only
d) The container itself

Answer: b

Q101. In Flexbox, what does align-items **align?**
a) Items along the main axis
b) Items along the cross axis
c) The container
d) Only the first item

Answer: b

Q102. Which CSS unit is relative to the font-size of the root element?
a) px
b) em
c) rem
d) pt

Answer: c

Q103. Which CSS unit is relative to the font-size of the parent element?
a) rem
b) em
c) vh
d) px

Answer: b

Q104. What does CSS Grid provide?
a) One-dimensional layout system
b) Two-dimensional layout system with rows and columns

c) Only column-based layouts
d) Only row-based layouts

Answer: b

Q105. How do you make a website mobile-friendly?
a) Use only desktop designs
b) Use responsive design techniques, flexible grids, and media queries
c) Create separate mobile sites only
d) Use smaller fonts only

Answer: b

Q106. What is a CSS preprocessor?
a) A tool that processes CSS before the browser sees it
b) A browser plugin
c) A JavaScript library
d) An HTML tag

Answer: a

Q107. Which of the following is a CSS preprocessor?
a) React
b) Sass
c) jQuery
d) Bootstrap

Answer: b

Q108. What is the purpose of CSS variables?
a) To store JavaScript values
b) To store reusable values in CSS
c) To create database queries
d) To define HTML attributes

Answer: b

Q109. How do you define a CSS variable?
a) $primary-color: blue;
b) @primary-color: blue;
c) --primary-color: blue;
d) var-primary-color: blue;

Answer: c

Q110. How do you use a CSS variable?
a) color: $primary-color;
b) color: @primary-color;
c) color: var(--primary-color);
d) color: primary-color;

Answer: c

Q111. What is the difference between display: none **and** visibility: hidden**?**
a) They are the same
b) display: none removes element from layout, visibility: hidden hides but occupies space
c) visibility: hidden removes element from layout
d) display: none hides but occupies space

Answer: b

Q112. What does the transition **property do in CSS?**
a) Changes element instantly
b) Creates smooth animation between property changes
c) Transitions between pages
d) Changes the display property

Answer: b

Q113. What does @keyframes **in CSS define?**
a) A JavaScript function
b) An animation sequence

c) A media query
d) A CSS variable

Answer: b

Q114. What is the difference between relative **and** absolute **units?**
a) Relative units are fixed, absolute units are flexible
b) Relative units are based on other values, absolute units are fixed
c) They are the same
d) Relative units are for fonts only

Answer: b

Q115. Which of the following is an absolute CSS unit?
a) em
b) rem
c) px
d) vw

Answer: c

Q116. Which of the following is a relative CSS unit?
a) cm
b) mm
c) in
d) vh

Answer: d

Q117. What does vw **unit stand for?**
a) Viewport width percentage
b) Variable width
c) Vertical width
d) Visual width

Answer: a

Q118. What does vh **unit stand for?**
a) Variable height
b) Viewport height percentage
c) Vertical height
d) Visual height

Answer: b

Q119. What is the purpose of box-sizing: border-box**?**
a) Includes padding and border in element's total width and height
b) Excludes padding from width calculation
c) Changes the box model completely
d) Adds a border to the box

Answer: a

Q120. What is a CSS framework?
a) A programming language
b) A pre-prepared library for styling web pages
c) A browser
d) A database

Answer: b

Q121. What is JavaScript?
a) A markup language
b) A styling language
c) A programming language for adding interactivity to web pages
d) A database language

Answer: c

Q122. Who invented JavaScript?
a) Tim Berners-Lee
b) Brendan Eich

c) Larry Page
d) Mark Zuckerberg

Answer: b

Q123. What is ECMAScript?
a) A different programming language
b) The standardized specification that JavaScript follows
c) A JavaScript library
d) A web browser

Answer: b

Q124. Which HTML tag is used to define internal JavaScript?
a) <javascript>
b) <js>
c) <script>
d) <code>

Answer: c

Q125. Where can JavaScript be placed in an HTML document?
a) Only in <head>
b) Only in <body>
c) In both <head> and <body>
d) Only in external files

Answer: c

Q126. How do you write "Hello World" in an alert box?
a) alert("Hello World");
b) msg("Hello World");
c) alertBox("Hello World");
d) popup("Hello World");

Answer: a

Q127. How do you create a function in JavaScript?
a) function myFunction()
b) create myFunction()
c) def myFunction()
d) func myFunction()

Answer: a

Q128. How do you call a function named "myFunction"?
a) call myFunction()
b) myFunction()
c) function myFunction()
d) run myFunction()

Answer: b

Q129. What is the correct way to write a comment in JavaScript?
a) <!-- comment -->
b) // comment
c) ' comment
d) -- comment

Answer: b

Q130. How do you write a multi-line comment in JavaScript?
a) // comment //
b) <!-- comment -->
c) /* comment */
d) ' comment '

Answer: c

Q131. Which operator is used to assign a value to a variable?
a) *
b) ==

c) =
d) ===

Answer: c

Q132. What is the correct way to declare a variable in modern JavaScript?
a) var name;
b) variable name;
c) let name;
d) v name;

Answer: c

Q133. What is the difference between let **and** const**?**
a) They are the same
b) let variables cannot be reassigned, const can
c) const variables cannot be reassigned, let can
d) let is for numbers only

Answer: c

Q134. What is a function in JavaScript?
a) A block of code designed to perform a particular task
b) A CSS selector
c) An HTML element
d) A database query

Answer: a

Q135. What is an array in JavaScript?
a) A function
b) A special variable that can hold multiple values
c) A CSS property
d) An HTML attribute

Answer: b

Q136. How do you access the first element of an array?
a) array[0]
b) array[1]
c) array.first()
d) array[first]

Answer: a

Q137. Which method adds an element to the end of an array?
a) array.add()
b) array.push()
c) array.append()
d) array.unshift()

Answer: b

Q138. Which method removes the last element from an array?
a) array.pop()
b) array.remove()
c) array.shift()
d) array.slice()

Answer: a

Q139. What is a JavaScript object?
a) A collection of key-value pairs
b) A function
c) An array
d) A CSS style

Answer: a

Q140. How do you access a property named "name" of an object "person"?
a) person[name]
b) person.name

c) person->name
d) person:name

Answer: b

Q141. What is the output of typeof 42**?**
a) "number"
b) "string"
c) "integer"
d) "float"

Answer: a

Q142. What is the output of typeof "hello"**?**
a) "string"
b) "text"
c) "word"
d) "char"

Answer: a

Q143. What is null **in JavaScript?**
a) An error
b) An intentional absence of any object value
c) 0
d) undefined

Answer: b

Q144. What is undefined **in JavaScript?**
a) A variable that has been declared but not assigned a value
b) An error
c) 0
d) false

Answer: a

Q145. Which operator checks for both value and type equality?
a) ==
b) =
c) ===
d) !=

Answer: c

Q146. What is the output of 2 == "2"**?**
a) true
b) false
c) error
d) undefined

Answer: a

Q147. What is the output of 2 === "2"**?**
a) true
b) false
c) error
d) undefined

Answer: b

Q148. What is a for loop used for?
a) To style elements
b) To execute a block of code multiple times
c) To create functions
d) To declare variables

Answer: b

Q149. What is the DOM in JavaScript?
a) Document Object Model - a representation of the HTML document
b) Data Object Model

c) Digital Output Method
d) Document Order Management

Answer: a

Q150. What does document **represent in JavaScript?**
a) A PDF file
b) The HTML document loaded in the browser
c) A text file
d) A CSS file

Answer: b

Q151. What is a closure in JavaScript?
a) A function that has access to its outer function's scope
b) A way to close a browser window
c) An HTML tag
d) A CSS property

Answer: a

Q152. What is the event loop in JavaScript?
a) A loop that runs events
b) A mechanism that handles asynchronous callbacks
c) A type of for loop
d) A DOM event

Answer: b

Q153. What is a Promise in JavaScript?
a) A guarantee that code will run
b) An object representing eventual completion of an async operation
c) A function that always runs
d) A synchronous operation

Answer: b

Q154. What does the async **keyword do?**
a) Makes a function run faster
b) Declares an asynchronous function
c) Creates a promise
d) Stops function execution

Answer: b

Q155. What does the await **keyword do?**
a) Pauses function execution until a promise settles
b) Waits for user input
c) Creates a delay
d) Stops all execution

Answer: a

Q156. What is JSON?
a) JavaScript Object Notation - a lightweight data interchange format
b) JavaScript Oriented Network
c) Java Source Object Name
d) JavaScript Object Navigator

Answer: a

Q157. Which method parses a JSON string into a JavaScript object?
a) JSON.stringify()
b) JSON.parse()
c) JSON.toObject()
d) JSON.convert()

Answer: b

Q158. Which method converts a JavaScript object to a JSON string?
a) JSON.parse()
b) JSON.toString()

c) JSON.stringify()
d) JSON.toJSON()

Answer: c

Q159. What is AJAX?
a) A cleaning product
b) Asynchronous JavaScript and XML - technique for updating parts of a page without reloading
c) A JavaScript library
d) A CSS framework

Answer: b

Q160. What is the Fetch API used for?
a) To fetch HTML elements
b) To make HTTP requests from JavaScript
c) To download files
d) To get CSS properties

Answer: b

Q161. Which method is used to add an event listener in JavaScript?
a) element.addEvent()
b) element.addEventListener()
c) element.on()
d) element.listen()

Answer: b

Q162. Which event occurs when a user clicks on an element?
a) mouseover
b) click
c) hover
d) onchange

Answer: b

Q163. What is event bubbling in JavaScript?
a) Events moving up through DOM hierarchy
b) Events that occur repeatedly
c) Creating bubbles in UI
d) A CSS animation

Answer: a

Q164. How do you prevent the default behavior of an event?
a) event.stop()
b) event.preventDefault()
c) event.cancel()
d) event.halt()

Answer: b

Q165. How do you stop event propagation?
a) event.stop()
b) event.preventDefault()
c) event.stopPropagation()
d) event.cancelBubble

Answer: c

Q166. What is the this **keyword in JavaScript?**
a) Refers to the current object/context
b) A reference to the previous object
c) A CSS selector
d) An HTML tag

Answer: a

Q167. What is prototypal inheritance in JavaScript?
a) Objects inheriting from other objects
b) Class-based inheritance

c) CSS inheritance
d) HTML parent-child relationship

Answer: a

Q168. What is a callback function?
a) A function passed as an argument to another function
b) A function that calls itself
c) A function that returns a number
d) A built-in function

Answer: a

Q169. What is callback hell?
a) A debugging error
b) Nested callbacks making code hard to read
c) A JavaScript runtime error
d) A browser crash

Answer: b

Q170. What is the spread operator in JavaScript?
a) ...
b) **
c) &&
d) ||

Answer: a

Q171. What does the map() **method do?**
a) Creates a new array with results of calling a function on every element
b) Maps coordinates
c) Creates a new object
d) Changes array size

Answer: a

Q172. What does the filter() **method do?**
a) Removes elements from array
b) Creates new array with elements that pass a test
c) Filters out undefined values
d) Changes all elements

Answer: b

Q173. What does the reduce() **method do?**
a) Reduces array size
b) Reduces array to a single value
c) Decreases numbers
d) Removes duplicates

Answer: b

Q174. What is destructuring in JavaScript?
a) Destroying objects
b) Extracting values from arrays or objects into distinct variables
c) Deleting properties
d) Breaking code

Answer: b

Q175. What is a module in JavaScript?
a) A CSS file
b) A reusable piece of code that can be exported and imported
c) An HTML component
d) A browser feature

Answer: b

Q176. What is a frontend framework?
a) A server-side technology
b) A structure for building user interfaces with pre-written code

c) A database system
d) A programming language

Answer: b

Q177. Which of the following is a popular JavaScript library for building user interfaces?
a) Django
b) Laravel
c) React
d) Flask

Answer: c

Q178. What is React developed by?
a) Google
b) Microsoft
c) Facebook (Meta)
d) Amazon

Answer: c

Q179. What is the virtual DOM in React?
a) A fake DOM
b) A lightweight copy of the real DOM for performance optimization
c) A browser feature
d) An HTML element

Answer: b

Q180. What is a component in React?
a) A CSS class
b) A reusable, independent piece of UI
c) An HTML page
d) A JavaScript variable

Answer: b

Q181. What is JSX in React?
a) A JavaScript extension that allows HTML-like syntax
b) A CSS preprocessor
c) A database query language
d) A testing tool

Answer: a

Q182. What is Angular developed by?
a) Facebook
b) Google
c) Microsoft
d) Apple

Answer: b

Q183. What is Vue.js?
a) A backend framework
b) A progressive JavaScript framework for building UIs
c) A database
d) A CSS library

Answer: b

Q184. What is state in frontend frameworks?
a) A CSS property
b) Data that determines how a component renders and behaves
c) A country
d) An HTML attribute

Answer: b

Q185. What are props in React?
a) Properties passed from parent to child components
b) A type of state

c) HTML attributes
d) CSS properties

Answer: a

Q186. What is Bootstrap?
a) A JavaScript framework
b) A CSS framework for responsive design
c) A backend technology
d) A database

Answer: b

Q187. What is Tailwind CSS?
a) A JavaScript library
b) A utility-first CSS framework
c) A backend framework
d) A database tool

Answer: b

Q188. What is jQuery?
a) A modern frontend framework
b) A fast, small JavaScript library for DOM manipulation
c) A CSS preprocessor
d) A server-side language

Answer: b

Q189. What is a single-page application (SPA)?
a) An application with only one page
b) A web app that loads a single HTML page and updates dynamically
c) A PDF document
d) A website with no navigation

Answer: b

Q190. Which of the following is NOT a frontend framework?
a) React
b) Vue
c) Angular
d) Express

Answer: d

Q191. What is backend development?
a) The visual part of a website
b) Server-side logic, databases, and application functionality
c) Only database management
d) Only API creation

Answer: b

Q192. Which language is commonly used for backend development?
a) HTML
b) CSS
c) Python
d) SCSS

Answer: c

Q193. What is a server?
a) A computer that serves requests to clients
b) A web browser
c) A programming language
d) A CSS framework

Answer: a

Q194. What is Node.js?
a) A frontend framework
b) A JavaScript runtime for server-side development

c) A database
d) A CSS library

Answer: b

Q195. What is an API?
a) Application Programming Interface
b) Advanced Program Integration
c) Application Protocol Interface
d) Automated Program Installation

Answer: a

Q196. What is a RESTful API?
a) An API that uses REST architectural style
b) An API that only gets data
c) An API that requires rest
d) A database API

Answer: a

Q197. What is Express.js?
a) A frontend framework
b) A web application framework for Node.js
c) A database
d) A CSS framework

Answer: b

Q198. What is Django?
a) A JavaScript library
b) A Python web framework
c) A database
d) A CSS preprocessor

Answer: b

Q199. What is Flask?
a) A lightweight Python web framework
b) A CSS framework
c) A database
d) A JavaScript library

Answer: a

Q200. What is Laravel?
a) A Python framework
b) A PHP framework
c) A JavaScript framework
d) A CSS framework

Answer: b

Q201. What is Ruby on Rails?
a) A JavaScript library
b) A web framework written in Ruby
c) A train simulation
d) A database

Answer: b

Q202. What is middleware in backend development?
a) Software that sits between the request and response
b) A database
c) A frontend component
d) A CSS property

Answer: a

Q203. What is server-side rendering?
a) Rendering pages in the browser
b) Generating HTML on the server and sending to client

c) Rendering images only
d) A CSS technique

Answer: b

Q204. What is a microservice architecture?
a) Building application as a collection of small, independent services
b) A single large application
c) A database design
d) A frontend pattern

Answer: a

Q205. What is serverless computing?
a) No servers involved
b) Running code without managing servers, using cloud functions
c) Disabled servers
d) Local development only

Answer: b

Q206. What is server-side programming?
a) Code that runs in the browser
b) Code that runs on the web server
c) Code that runs in databases
d) Code that styles pages

Answer: b

Q207. What is PHP?
a) A client-side language
b) A server-side scripting language
c) A CSS framework
d) A database

Answer: b

Q208. What is Python commonly used for in web development?
a) Only frontend
b) Server-side logic with frameworks like Django and Flask
c) Only data science
d) Styling web pages

Answer: b

Q209. What is JSP?
a) JavaScript Pages
b) JavaServer Pages
c) Java Styling Pages
d) JavaScript Programming

Answer: b

Q210. What is CGI in web development?
a) Common Gateway Interface
b) Computer Graphics Interface
c) Code Generation Interface
d) Client Gateway Integration

Answer: b

Q211. What is session management in web applications?
a) Managing user meetings
b) Maintaining user state across multiple requests
c) Database management
d) Code organization

Answer: b

Q212. How is session information typically maintained?
a) Only in URLs
b) Using cookies, session IDs, or tokens

c) In HTML comments
d) In CSS

Answer: b

Q213. What is authentication in web applications?
a) Verifying user identity
b) Determining user permissions
c) Managing user data
d) Styling login forms

Answer: a

Q214. What is authorization in web applications?
a) Verifying identity
b) Determining what resources a user can access
c) Creating user accounts
d) Logging user activities

Answer: b

Q215. What is a web framework?
a) A structure for building web applications
b) A type of database
c) A CSS library
d) A browser

Answer: a

Q216. What is routing in web applications?
a) Mapping URLs to specific handlers or functions
b) Network routing
c) Database routing
d) CSS path

Answer: a

Q217. What is a template engine?
a) A tool for designing HTML templates
b) A tool that combines templates with data to generate HTML
c) A CSS preprocessor
d) A JavaScript library

Answer: b

Q218. What is Jinja2?
a) A JavaScript framework
b) A template engine for Python
c) A CSS framework
d) A database

Answer: b

Q219. What is EJS?
a) Embedded JavaScript templating
b) Enhanced JavaScript
c) External JavaScript
d) Encoded JavaScript

Answer: a

Q220. What is a controller in MVC architecture?
a) Handles user input and interacts with the model
b) Manages database connections
c) Renders the view
d) Stores data

Answer: a

Q221. What is a model in MVC architecture?
a) Manages data and business logic
b) Handles user interface

c) Processes HTTP requests
d) Routes URLs

Answer: a

Q222. What is a view in MVC architecture?
a) Handles data logic
b) Presents data to the user
c) Processes input
d) Manages sessions

Answer: b

Q223. What is ORM in backend development?
a) Object-Relational Mapping
b) Object Request Model
c) Online Resource Management
d) Open Resource Mapping

Answer: a

Q224. What is the purpose of an ORM?
a) To map objects to database tables
b) To create HTML
c) To style pages
d) To manage sessions

Answer: a

Q225. What is SQLAlchemy?
a) A JavaScript library
b) An ORM for Python
c) A CSS framework
d) A database

Answer: b

Q226. What is a database?
a) A structured collection of data
b) A programming language
c) A web server
d) A browser

Answer: a

Q227. What is a relational database?
a) Database with tables related to each other
b) Database with no relationships
c) Database that stores documents
d) Database that stores key-value pairs

Answer: a

Q228. Which language is used to query relational databases?
a) HTML
b) CSS
c) SQL
d) JavaScript

Answer: c

Q229. What does SQL stand for?
a) Structured Query Language
b) Simple Query Language
c) Sequential Query Language
d) Standard Query Language

Answer: a

Q230. Which of the following is a relational database?
a) MongoDB
b) MySQL

c) Redis
d) Cassandra

Answer: b

Q231. Which of the following is a NoSQL database?
a) PostgreSQL
b) MySQL
c) MongoDB
d) Oracle

Answer: c

Q232. What is MongoDB?
a) A relational database
b) A NoSQL document database
c) A programming language
d) A web server

Answer: b

Q233. What is a primary key in a database?
a) A unique identifier for each record in a table
b) The first column in a table
c) A key to open the database
d) An index

Answer: a

Q234. What is a foreign key?
a) A key from another country
b) A field that links to the primary key of another table
c) A duplicate key
d) A backup key

Answer: b

Q235. What SQL command is used to retrieve data?
a) INSERT
b) UPDATE
c) SELECT
d) DELETE

Answer: c

Q236. What SQL command is used to add new data?
a) SELECT
b) ADD
c) INSERT
d) CREATE

Answer: c

Q237. What SQL command is used to modify existing data?
a) UPDATE
b) CHANGE
c) MODIFY
d) ALTER

Answer: a

Q238. What SQL command is used to remove data?
a) REMOVE
b) DELETE
c) DROP
d) CLEAR

Answer: b

Q239. What is a JOIN in SQL?
a) Combining rows from two or more tables based on related columns
b) Adding two numbers

c) Connecting to database
d) Merging databases

Answer: a

Q240. What is database indexing?
a) Creating a table of contents for faster data retrieval
b) Listing all databases
c) Indexing web pages
d) Numbering rows

Answer: a

Q241. What is ACID in databases?
a) Atomicity, Consistency, Isolation, Durability
b) A type of database
c) A SQL command
d) A database model

Answer: a

Q242. What is a database connection pool?
a) A pool of water
b) A cache of database connections for reuse
c) A type of database
d) A connection string

Answer: b

Q243. What is SQL injection?
a) Injecting SQL code into an application to attack the database
b) Inserting data quickly
c) A database backup method
d) A type of JOIN

Answer: a

Q244. How can SQL injection be prevented?
a) Using more SQL
b) Using parameterized queries and input validation
c) Disabling the database
d) Using complex passwords only

Answer: b

Q245. What is database migration?
a) Moving database to another server
b) Managing and applying changes to database schema
c) Moving data between tables
d) Database backup

Answer: b

Q246. What is the OWASP Top Ten?
a) A list of top ten web servers
b) A list of top ten security risks for web applications
c) A top ten programming languages list
d) A top ten frameworks list

Answer: b

Q247. What is SQL Injection?
a) A way to optimize SQL queries
b) An attack where malicious SQL code is inserted into input fields
c) A database feature
d) A SQL command

Answer: b

Q248. What is Cross-Site Scripting (XSS)?
a) A way to link websites
b) An attack where malicious scripts are injected into trusted websites

c) A browser feature
d) A CSS technique

Answer: b

Q249. What is the difference between stored and reflected XSS?
a) Stored is saved on server, reflected is in request/response
b) Reflected is more dangerous
c) Stored affects only one user
d) They are the same

Answer: a

Q250. What is Cross-Site Request Forgery (CSRF)?
a) Forging requests to another site
b) An attack that tricks user into executing unwanted actions
c) A way to copy websites
d) A security feature

Answer: b

Q251. How does CSRF typically work?
a) By injecting scripts
b) By using the user's authenticated session to perform unauthorized actions
c) By stealing passwords
d) By hacking the server

Answer: b

Q252. What is broken authentication?
a) A feature
b) Weaknesses in authentication mechanisms that allow attackers to compromise credentials
c) A type of encryption
d) A login form

Answer: b

Q253. What is sensitive data exposure?
a) Sharing data publicly
b) When sensitive information is not properly protected
c) A backup method
d) Data encryption

Answer: b

Q254. What is security misconfiguration?
a) Proper security setup
b) Using insecure default configurations or incomplete configurations
c) A security feature
d) A firewall setting

Answer: b

Q255. What is broken access control?
a) Users accessing resources they shouldn't be allowed to
b) Proper permission system
c) A login system
d) Password protection

Answer: a

Q256. What is HTTPS?
a) HTTP with encryption
b) A faster version of HTTP
c) HTTP without security
d) A different protocol

Answer: a

Q257. What does SSL/TLS provide?
a) Faster browsing
b) Encryption, authentication, and data integrity

c) More storage
d) Better images

Answer: b

Q258. What is an SSL certificate?
a) A digital certificate that authenticates a website's identity and enables encryption
b) A type of virus
c) A browser plugin
d) A programming certificate

Answer: a

Q259. What is input validation?
a) Checking user input for correctness and safety
b) Entering data
c) Validating forms with JavaScript only
d) A database feature

Answer: a

Q260. What is the difference between whitelist and blacklist validation?
a) Whitelist allows only known good, blacklist blocks known bad
b) They are the same
c) Blacklist is better
d) Whitelist is for colors only

Answer: a

Q261. What is output encoding?
a) Compressing output
b) Converting data to a safe format before displaying
c) Encrypting output
d) Minifying output

Answer: b

Q262. What is a Content Security Policy (CSP)?
a) A security standard to prevent XSS and other injection attacks
b) A content management system
c) A CSS property
d) A JavaScript framework

Answer: a

Q263. What is CORS?
a) Cross-Origin Resource Sharing
b) Cross-Origin Request Security
c) Cross-Site Resource Sharing
d) Cross-Origin Response System

Answer: a

Q264. What is session hijacking?
a) Taking over a valid user session
b) Creating a new session
c) Ending a session
d) Saving a session

Answer: a

Q265. How can session hijacking be prevented?
a) Using secure cookies, session timeouts, and HTTPS
b) Using longer session IDs only
c) Not using sessions
d) Using simple session IDs

Answer: a

Q266. What is a man-in-the-middle attack?
a) Attacker intercepts communication between two parties
b) Attack on middle servers

c) A type of virus
d) Network maintenance

Answer: a

Q267. What is HTTPS redirection?
a) Redirecting HTTP to HTTPS
b) Redirecting to another site
c) A 301 redirect
d) A meta refresh

Answer: a

Q268. What is secure password storage practice?
a) Storing passwords in plain text
b) Hashing passwords with salt
c) Encrypting passwords only
d) Using short passwords

Answer: b

Q269. What is two-factor authentication (2FA)?
a) Using two passwords
b) Using two different methods to verify identity
c) Logging in twice
d) Two-step login form

Answer: b

Q270. What is a security header in HTTP?
a) HTTP headers that enhance security
b) A secure connection
c) A firewall rule
d) An encryption method

Answer: a

Q271. What is web performance optimization?
a) Making websites look better
b) Techniques to make websites load faster
c) Adding more features
d) Using more images

Answer: b

Q272. What is minification in web development?
a) Making fonts smaller
b) Removing unnecessary characters from code without changing functionality
c) Reducing image size
d) Using smaller screens

Answer: b

Q273. What is code bundling?
a) Combining multiple files into fewer files
b) Writing more code
c) Adding comments
d) Creating packages

Answer: a

Q274. What is lazy loading?
a) Loading content only when needed
b) Loading everything at once
c) Slow loading
d) Not loading anything

Answer: a

Q275. What is browser caching?
a) Storing website resources locally to avoid re-downloading
b) Deleting browser history

c) Clearing cookies
d) Saving passwords

Answer: a

Q276. What is a CDN?
a) Content Delivery Network
b) Code Distribution Network
c) Computer Data Network
d) Content Download Network

Answer: a

Q277. How does a CDN improve performance?
a) By serving content from servers closer to users
b) By compressing images
c) By minifying code
d) By using faster computers

Answer: a

Q278. What is image optimization?
a) Making images look better
b) Reducing image file size without significant quality loss
c) Adding more images
d) Removing images

Answer: b

Q279. What is critical rendering path?
a) The path to CSS files
b) Steps browser takes to render a page
c) A JavaScript path
d) Network route

Answer: b

Q280. What is render-blocking resource?
a) Resources that prevent page from rendering quickly
b) Resources that display content
c) CSS only
d) JavaScript only

Answer: a

Q281. What is tree shaking in modern JavaScript?
a) Removing dead code from bundles
b) Removing trees from images
c) CSS optimization
d) HTML cleanup

Answer: a

Q282. What is a performance budget?
a) Money for performance
b) A limit on page size or load time
c) Budget for hosting
d) Developer salary

Answer: b

Q283. What is the purpose of async and defer attributes on script tags?
a) To make scripts run faster
b) To control when scripts load and execute
c) To hide scripts
d) To encrypt scripts

Answer: b

Q284. What is the difference between async and defer?
a) They are the same
b) async loads independently, defer loads in order after HTML parsing

c) async is faster
d) defer is older

Answer: b

Q285. What is a Cumulative Layout Shift (CLS) in web performance?
a) Changing layouts
b) A metric measuring visual stability
c) A CSS property
d) A JavaScript function

Answer: b

Q286. What is unit testing?
a) Testing entire application
b) Testing individual components or functions
c) Testing units of measurement
d) Testing user interface

Answer: b

Q287. What is integration testing?
a) Testing how different modules work together
b) Testing single components
c) Testing UI only
d) Testing databases only

Answer: a

Q288. What is end-to-end testing?
a) Testing from start to finish of application flow
b) Testing only the end
c) Testing APIs only
d) Testing endpoints only

Answer: a

Q289. What is a testing framework?
a) A tool for writing and running tests
b) A CSS framework
c) A JavaScript library
d) A database tool

Answer: a

Q290. Which of the following is a JavaScript testing framework?
a) Django
b) Jest
c) Laravel
d) Flask

Answer: b

Q291. What is debugging?
a) Writing code
b) Finding and fixing errors in code
c) Testing code
d) Optimizing code

Answer: b

Q292. What are browser developer tools?
a) Tools built into browsers for debugging and testing
b) External software
c) Code editors
d) Database tools

Answer: a

Q293. What is console.log() used for?
a) Logging into applications
b) Printing debug information to the console

c) Creating log files
d) Sending logs to server

Answer: b

Q294. What is a breakpoint in debugging?
a) A point where code execution pauses for inspection
b) A broken point in code
c) A CSS property
d) An HTML tag

Answer: a

Q295. What is manual testing?
a) Testing done by humans without automation
b) Testing with machines
c) Testing manuals
d) Writing test cases

Answer: a

Q296. What is automated testing?
a) Testing with automation tools
b) Manual testing
c) Testing automatically
d) No testing

Answer: a

Q297. What is a test case?
a) A set of conditions to determine if a feature works correctly
b) A computer case
c) A test environment
d) A code file

Answer: a

Q298. What is test-driven development (TDD)?
a) Writing tests after code
b) Writing tests before code, then writing code to pass tests
c) Testing during development
d) No testing approach

Answer: b

Q299. What is regression testing?
a) Testing new features only
b) Testing to ensure new changes haven't broken existing functionality
c) Testing old versions
d) Testing databases

Answer: b

Q300. What is the purpose of linters in development?
a) To find and fix code style issues and potential errors
b) To run tests
c) To compile code
d) To minify code

Answer: a

Q301. What is web hosting?
a) A service that makes websites accessible on the internet
b) Hosting parties
c) A programming service
d) A database service

Answer: a

Q302. What is a domain name?
a) The address of a website (e.g., example.com)
b) A category of websites

c) A server name
d) An IP address

Answer: a

Q303. What is DNS?
a) Domain Name System - translates domain names to IP addresses
b) Digital Network Service
c) Domain Network Security
d) Data Name System

Answer: a

Q304. What is deployment in web development?
a) Making code available for use on a server
b) Writing code
c) Testing code
d) Deleting code

Answer: a

Q305. What is FTP used for?
a) File Transfer Protocol - uploading files to servers
b) Fast Transfer Protocol
c) File Testing Protocol
d) Frontend Transfer Program

Answer: a

Q306. What is continuous integration?
a) Regularly merging code changes into a shared repository
b) Continuous testing
c) Continuous deployment
d) Writing code continuously

Answer: a

Q307. What is continuous deployment?
a) Automatically deploying code changes to production
b) Manual deployment
c) Testing deployment
d) Planning deployment

Answer: a

Q308. What is a staging environment?
a) An environment that mirrors production for final testing
b) The production server
c) Development machine
d) A testing tool

Answer: a

Q309. What is version control?
a) Managing changes to code over time
b) Controlling versions of software
c) A deployment tool
d) A testing framework

Answer: a

Q310. What is Git?
a) A distributed version control system
b) A programming language
c) A hosting service
d) A database

Answer: a

Q311. What is a Web API?
a) An API designed for web applications
b) A web page

c) A database
d) A server

Answer: a

Q312. What is a REST API?
a) An API that follows REST architectural constraints
b) An API that requires rest
c) A database API
d) A local API

Answer: a

Q313. What is GraphQL?
a) A query language for APIs
b) A database
c) A graph database
d) A CSS framework

Answer: a

Q314. What is the difference between REST and GraphQL?
a) REST has multiple endpoints, GraphQL typically has one endpoint with flexible queries
b) They are the same
c) GraphQL is older
d) REST is newer

Answer: a

Q315. What is an API endpoint?
a) A URL where an API can be accessed
b) The end of an API
c) A server
d) A database connection

Answer: a

Q316. What is authentication in APIs?
a) Verifying the identity of the API client
b) Authorizing access
c) API design
d) API testing

Answer: a

Q317. What is an API key?
a) A unique identifier for API authentication
b) A key to open APIs
c) A database key
d) An encryption key

Answer: a

Q318. What is OAuth?
a) An open standard for access delegation
b) A programming language
c) A database
d) An API

Answer: a

Q319. What is JWT?
a) JSON Web Token
b) JavaScript Web Tool
c) Java Web Technology
d) JSON Web Transfer

Answer: a

Q320. What is webhook?
a) An HTTP callback that occurs when something happens
b) A hook in the web

c) A JavaScript function
d) An API endpoint

Answer: a

Q321. What is Jamstack?
a) JavaScript, APIs, and Markup architecture
b) A stack of jams
c) A programming language
d) A database

Answer: a

Q322. What is a headless CMS?
a) A content management system without the frontend
b) A CMS with no content
c) A database
d) A framework

Answer: a

Q323. What is server-side rendering (SSR)?
a) Rendering pages on the server
b) Rendering in browser
c) Rendering images
d) Rendering only

Answer: a

Q324. What is static site generation (SSG)?
a) Generating HTML pages at build time
b) Generating pages at request time
c) Static pages only
d) No generation

Answer: a

Q325. What is incremental static regeneration?
a) Updating static content without rebuilding entire site
b) Regenerating all pages
c) Static only
d) Dynamic only

Answer: a

Q326. What is PWA?
a) Progressive Web Application
b) Professional Web App
c) Progressive Web Architecture
d) Primary Web Application

Answer: a

Q327. What features do PWAs typically have?
a) Offline capability, push notifications, installable
b) Only online
c) No special features
d) Desktop only

Answer: a

Q328. What is a service worker?
a) A script that runs in the background for offline features
b) A web worker
c) A server worker
d) A database worker

Answer: a

Q329. What is JAMstack?
a) Modern web architecture using JavaScript, APIs, and Markup
b) A framework

c) A library
d) A database

Answer: a

Q330. What is the benefit of modern build tools?
a) Automating tasks like bundling, minification, and transpilation
b) Slowing development
c) Removing features
d) Adding complexity

Answer: a